

Gabriel Akurang 35882028 Cohort A

```
MariaDB [(none)]> create database FACT;  
Query OK, 1 row affected (0.007 sec)
```

```
Takyi@Akurang-Pro ~ % sed -i ' ' 's/GETDATE()/CURDATE()/g'  
/Users/Takyi/Downloads/Fact_SQL-1.sql
```

```
Takyi@Akurang-Pro ~ % perl -pi -e  
"s/'(\d{2})\\/'(\d{2})\\/'(\d{4})'/'\3-\1-\2'/g"  
/Users/Takyi/Downloads/Fact_SQL-1.sql
```

```
Takyi@Akurang-Pro ~ % mariadb FACT < /Users/Takyi/Downloads/Fact_SQL-1.sql;
```

```
MariaDB [(none)]> use FACT;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A
```

Database changed

#1

```
MariaDB [FACT]> select book_title, book_cost, book_year  
-> from book;
```

```
+-----+-----+  
| book_title | book_cost | book_year |  
+-----+-----+  
| Beginner's Guide to JAVA | 59.95 | 2012 |  
| Database in the Cloud | 79.95 | 2012 |  
| Mastering the database environment | 89.95 | 2013 |  
| Conceptual Programming | 59.95 | 2013 |  
| J++ in Mobile Apps | 49.95 | 2013 |  
| iOS Programming | 79.95 | 2013 |  
| JAVA First Steps | 49.95 | 2013 |  
| C# in Middleware Deployment | 59.95 | 2013 |  
| DATABASES in Theory | 129.95 | 2013 |  
| Cloud-based Mobile Applications | 69.95 | 2013 |  
| The Golden Road to Platform independence | 119.95 | 2014 |  
| Capture the Cloud | 69.95 | 2014 |  
| Shining Through the Cloud: Sun Programming | 109.95 | 2014 |  
| What You Always Wanted to Know About Database, But Were Afraid to Ask | 49.95 | 2014 |  
| Starlight Applications | 69.95 | 2014 |  
| Reengineering the Middle Tier | 89.95 | 2014 |  
| Thoughts on Revitalizing Ruby | 59.95 | 2014 |  
| Beyond the Database Veil |
```

```

69.95 |      2014 |
| Virtual Programming for Virtual Environments |
79.95 |      2014 |
| Coding Style for Maintenance |
49.95 |      2015 |
+-----+
+-----+
20 rows in set (0.002 sec)

```

#2

```

MariaDB [FACT]> select pat_fname, pat_lname
-> from patron;

```

```

+-----+-----+
| pat_fname | pat_lname |
+-----+-----+
| robert    | carter    |
| Kelsey    | Koch      |
| Cedric    | Baldwin   |
| Vera      | Alvarado  |
| Alan      | Martin    |
| Cory      | Barry     |
| Peggy     | Marsh     |
| Tony      | Miles     |
| Betsy     | Malone    |
| Nadine    | Blair     |
| Allen     | Horne     |
| Jamal     | Melendez  |
| Helena    | Hughes    |
| Jimmie    | Love      |
| Sandra    | Yang      |
| Lorenzo   | Torres    |
| Shelby    | Noble     |
| Holly     | Anthony   |
| Tyler     | Pope      |
| Thomas    | Duran     |
| Claire    | Gomez     |
| Iva       | Ramos     |
| Ollie     | Cantrell  |
| Rena      | Mathis    |
| Keith     | Cooley    |
| Jerald    | Gaines    |
| Iva       | McClain   |
| Desiree   | Rivas     |
| Marina    | King      |
| Maureen   | Downs     |
| Angel     | Terrell   |
| Desiree   | Harrington|
| Carlton  | Morton    |
| Gloria    | Pitts     |
| Zach      | Kelly     |
| Jose      | Hays      |
| Jewel     | England   |
| Wilfred   | Fuller    |
| Jeff      | Owens     |
| Alicia    | Dickson   |
| Homer     | Goodman   |
| Gerald    | Burke     |
| Brandi    | Larson    |
| Erika     | Bowen     |
| Elton     | Irwin     |
| Jan       | Joyce     |
| Irene     | West      |
| Mario     | King      |
| Roberto   | Kennedy   |

```

```
| Leon      | Richmond  |
+-----+-----+
50 rows in set (0.001 sec)
```

#3

```
MariaDB [FACT]> select check_num, check_out_date, check_due_date
-> from checkout;
```

check_num	check_out_date	check_due_date
91001	2015-03-31	2015-04-14
91002	2015-03-31	2015-04-07
91003	2015-03-31	2015-04-14
91004	2015-03-31	2015-04-14
91005	2015-03-31	2015-04-07
91006	2015-04-05	2015-04-12
91007	2015-04-05	2015-04-12
91008	2015-04-05	2015-04-12
91009	2015-04-05	2015-04-19
91010	2015-04-05	2015-04-19
91011	2015-04-05	2015-04-12
91012	2015-04-08	2015-04-15
91013	2015-04-10	2015-04-24
91014	2015-04-11	2015-04-18
91015	2015-04-11	2015-04-18
91016	2015-04-13	2015-04-27
91017	2015-04-14	2015-04-21
91018	2015-04-14	2015-04-28
91019	2015-04-14	2015-04-21
91020	2015-04-16	2015-04-23
91021	2015-04-16	2015-04-23
91022	2015-04-16	2015-04-23
91023	2015-04-16	2015-04-23
91024	2015-04-21	2015-04-28
91025	2015-04-21	2015-04-28
91026	2015-04-21	2015-05-04
91027	2015-04-21	2015-05-04
91028	2015-04-22	2015-05-05
91029	2015-04-22	2015-04-29
91030	2015-04-22	2015-04-29
91031	2015-04-23	2015-05-06
91032	2015-04-23	2015-04-30
91033	2015-04-23	2015-04-30
91034	2015-04-23	2015-04-30
91035	2015-04-24	2015-05-07
91036	2015-04-24	2015-04-30
91037	2015-04-28	2015-05-04
91038	2015-04-28	2015-05-04
91039	2015-04-28	2015-05-04
91040	2015-04-28	2015-05-04
91041	2015-04-28	2015-05-04
91042	2015-04-29	2015-05-05
91043	2015-04-29	2015-05-05
91044	2015-04-30	2015-05-07
91045	2015-04-30	2015-05-07
91046	2015-04-30	2015-05-07
91047	2015-04-30	2015-05-07
91048	2015-05-04	2015-05-11
91049	2015-05-04	2015-05-11
91050	2015-05-08	2015-05-15
91051	2015-05-08	2015-05-15
91052	2015-05-08	2015-05-15
91053	2015-05-09	2015-05-16
91054	2015-05-09	2015-05-16

91055	2015-05-09	2015-05-16
91056	2015-05-10	2015-05-17
91057	2015-05-10	2015-05-17
91058	2015-05-10	2015-05-17
91059	2015-05-10	2015-05-17
91060	2015-05-15	2015-05-22
91061	2015-05-15	2015-05-22
91062	2015-05-15	2015-05-22
91063	2015-05-15	2015-05-22
91064	2015-05-17	2015-05-31
91065	2015-05-17	2015-05-24
91066	2015-05-19	2015-05-26
91067	2015-05-24	2015-05-31
91068	2015-05-24	2015-05-31

68 rows in set (0.001 sec)

#4

MariaDB [FACT]> select book_num, book_title, book_year from book;

book_num	book_title	book_year
5235	Beginner's Guide to JAVA	2012
5236	Database in the Cloud	2012
5237	Mastering the database environment	2013
5238	Conceptual Programming	2013
5239	J++ in Mobile Apps	2013
5240	iOS Programming	2013
5241	JAVA First Steps	2013
5242	C# in Middleware Deployment	2013
5243	DATABASES in Theory	2013
5244	Cloud-based Mobile Applications	2013
5245	The Golden Road to Platform independence	2014
5246	Capture the Cloud	2014
5247	Shining Through the Cloud: Sun Programming	2014
5248	What You Always Wanted to Know About Database, But Were Afraid to	2014
5249	Starlight Applications	2014
5250	Reengineering the Middle Tier	2014
5251	Thoughts on Revitalizing Ruby	2014
5252	Beyond the Database Veil	2014

```

|      5253 | Virtual Programming for Virtual Environments
|      2014 |
|      5254 | Coding Style for Maintenance
|      2015 |
+-----+
+-----+
+-----+
20 rows in set (0.000 sec)

```

#5

```

MariaDB [FACT]> select distinct book_year from book;
+-----+
| book_year |
+-----+
|      2012 |
|      2013 |
|      2014 |
|      2015 |
+-----+
4 rows in set (0.002 sec)

```

#6

```

MariaDB [FACT]> select distinct book_subject from book;
+-----+
| book_subject |
+-----+
| Programming  |
| Cloud        |
| Database      |
| Middleware    |
+-----+
4 rows in set (0.000 sec)

```

#7

```

MariaDB [FACT]> select book_num, book_title, book_cost from book;
+-----+
+-----+
+-----+
| book_num | book_title
| book_cost |
+-----+
+-----+
|      5235 | Beginner's Guide to JAVA
|      59.95 |
|      5236 | Database in the Cloud
|      79.95 |
|      5237 | Mastering the database environment
|      89.95 |
|      5238 | Conceptual Programming
|      59.95 |
|      5239 | J++ in Mobile Apps
|      49.95 |
|      5240 | iOS Programming
|      79.95 |
|      5241 | JAVA First Steps
|      49.95 |
|      5242 | C# in Middleware Deployment
|      59.95 |
|      5243 | DATABASES in Theory

```

```

| 129.95 |
| 5244 | Cloud-based Mobile Applications
| 69.95 |
| 5245 | The Golden Road to Platform independence
| 119.95 |
| 5246 | Capture the Cloud
| 69.95 |
| 5247 | Shining Through the Cloud: Sun Programming
| 109.95 |
| 5248 | What You Always Wanted to Know About Database, But Were Afraid to
Ask | 49.95 |
| 5249 | Starlight Applications
| 69.95 |
| 5250 | Reengineering the Middle Tier
| 89.95 |
| 5251 | Thoughts on Revitalizing Ruby
| 59.95 |
| 5252 | Beyond the Database Veil
| 69.95 |
| 5253 | Virtual Programming for Virtual Environments
| 79.95 |
| 5254 | Coding Style for Maintenance
| 49.95 |
+-----+
+-----+
+-----+
20 rows in set (0.000 sec)

```

#8

```

MariaDB [FACT]> select check_num, book_num, pat_id, check_out_date,
check_due_date from checkout order by check_out_date desc;

```

```

+-----+-----+-----+-----+-----+
| check_num | book_num | pat_id | check_out_date | check_due_date |
+-----+-----+-----+-----+-----+
| 91067 | 5252 | 1229 | 2015-05-24 | 2015-05-31 |
| 91068 | 5238 | 1229 | 2015-05-24 | 2015-05-31 |
| 91066 | 5242 | 1228 | 2015-05-19 | 2015-05-26 |
| 91065 | 5244 | 1210 | 2015-05-17 | 2015-05-24 |
| 91064 | 5236 | 1183 | 2015-05-17 | 2015-05-31 |
| 91060 | 5235 | 1209 | 2015-05-15 | 2015-05-22 |
| 91061 | 5246 | 1172 | 2015-05-15 | 2015-05-22 |
| 91062 | 5254 | 1223 | 2015-05-15 | 2015-05-22 |
| 91063 | 5243 | 1223 | 2015-05-15 | 2015-05-22 |
| 91056 | 5254 | 1224 | 2015-05-10 | 2015-05-17 |
| 91057 | 5238 | 1224 | 2015-05-10 | 2015-05-17 |
| 91058 | 5252 | 1171 | 2015-05-10 | 2015-05-17 |
| 91059 | 5249 | 1207 | 2015-05-10 | 2015-05-17 |
| 91055 | 5246 | 1221 | 2015-05-09 | 2015-05-16 |
| 91054 | 5236 | 1221 | 2015-05-09 | 2015-05-16 |
| 91053 | 5240 | 1212 | 2015-05-09 | 2015-05-16 |
| 91052 | 5236 | 1213 | 2015-05-08 | 2015-05-15 |
| 91050 | 5236 | 1220 | 2015-05-08 | 2015-05-15 |
| 91051 | 5237 | 1222 | 2015-05-08 | 2015-05-15 |
| 91049 | 5240 | 1214 | 2015-05-04 | 2015-05-11 |
| 91048 | 5249 | 1229 | 2015-05-04 | 2015-05-11 |
| 91047 | 5236 | 1218 | 2015-04-30 | 2015-05-07 |
| 91046 | 5235 | 1225 | 2015-04-30 | 2015-05-07 |
| 91045 | 5242 | 1210 | 2015-04-30 | 2015-05-07 |
| 91044 | 5248 | 1219 | 2015-04-30 | 2015-05-07 |
| 91043 | 5244 | 1226 | 2015-04-29 | 2015-05-05 |
| 91042 | 5235 | 1220 | 2015-04-29 | 2015-05-05 |
| 91041 | 5236 | 1211 | 2015-04-28 | 2015-05-04 |

```

91040	5237	1228	2015-04-28	2015-05-04
91039	5240	1222	2015-04-28	2015-05-04
91038	5238	1215	2015-04-28	2015-05-04
91037	5235	1210	2015-04-28	2015-05-04
91036	5237	1202	2015-04-24	2015-04-30
91035	5248	1165	2015-04-24	2015-05-07
91034	5240	1185	2015-04-23	2015-04-30
91033	5235	1174	2015-04-23	2015-04-30
91032	5238	1172	2015-04-23	2015-04-30
91031	5252	1165	2015-04-23	2015-05-06
91030	5242	1207	2015-04-22	2015-04-29
91029	5244	1203	2015-04-22	2015-04-29
91028	5236	1183	2015-04-22	2015-05-05
91024	5235	1209	2015-04-21	2015-04-28
91025	5246	1172	2015-04-21	2015-04-28
91027	5243	1161	2015-04-21	2015-05-04
91026	5254	1161	2015-04-21	2015-05-04
91023	5249	1207	2015-04-16	2015-04-23
91022	5252	1171	2015-04-16	2015-04-23
91021	5238	1185	2015-04-16	2015-04-23
91020	5254	1185	2015-04-16	2015-04-23
91017	5240	1184	2015-04-14	2015-04-21
91018	5236	1170	2015-04-14	2015-04-28
91019	5246	1171	2015-04-14	2015-04-21
91016	5236	1183	2015-04-13	2015-04-27
91015	5237	1204	2015-04-11	2015-04-18
91014	5236	1204	2015-04-11	2015-04-18
91013	5240	1160	2015-04-10	2015-04-24
91012	5249	1181	2015-04-08	2015-04-15
91006	5235	1203	2015-04-05	2015-04-12
91007	5244	1174	2015-04-05	2015-04-12
91008	5248	1181	2015-04-05	2015-04-12
91009	5242	1170	2015-04-05	2015-04-19
91010	5235	1161	2015-04-05	2015-04-19
91011	5236	1181	2015-04-05	2015-04-12
91005	5236	1202	2015-03-31	2015-04-07
91004	5237	1160	2015-03-31	2015-04-14
91003	5240	1160	2015-03-31	2015-04-14
91002	5238	1209	2015-03-31	2015-04-07
91001	5235	1165	2015-03-31	2015-04-14

68 rows in set (0.001 sec)

#9

MariaDB [FACT]> select book_title, book_year, book_subject from book order by book_subject asc, book_year desc, book_title asc;

book_title	
book_year book_subject	
Capture the Cloud	
2014 Cloud	
Starlight Applications	
2014 Cloud	
Cloud-based Mobile Applications	
2013 Cloud	
Database in the Cloud	
2012 Cloud	
Beyond the Database Veil	
2014 Database	
What You Always Wanted to Know About Database, But Were Afraid to Ask	

```

2014 | Database |
| DATABASES in Theory |
2013 | Database |
| Mastering the database environment |
2013 | Database |
| Reengineering the Middle Tier |
2014 | Middleware |
| The Golden Road to Platform independence |
2014 | Middleware |
| C# in Middleware Deployment |
2013 | Middleware |
| Coding Style for Maintenance |
2015 | Programming |
| Shining Through the Cloud: Sun Programming |
2014 | Programming |
| Thoughts on Revitalizing Ruby |
2014 | Programming |
| Virtual Programming for Virtual Environments |
2014 | Programming |
| Conceptual Programming |
2013 | Programming |
| iOS Programming |
2013 | Programming |
| J++ in Mobile Apps |
2013 | Programming |
| JAVA First Steps |
2013 | Programming |
| Beginner's Guide to JAVA |
2012 | Programming |
+-----+
+-----+-----+
20 rows in set (0.000 sec)

```

#10

```

MariaDB [FACT]> select book_num, book_title, book_year
-> from book
-> where book_year = 2012;
+-----+-----+-----+
| book_num | book_title | book_year |
+-----+-----+-----+
| 5235 | Beginner's Guide to JAVA | 2012 |
| 5236 | Database in the Cloud | 2012 |
+-----+-----+-----+
2 rows in set (0.002 sec)

```

#11

```

MariaDB [FACT]> select book_num, book_title, book_year from book where
book_title like '%database%';
+-----+
+-----+
+-----+
| book_num | book_title |
| book_year |
+-----+
+-----+
| 5236 | Database in the Cloud |
| 2012 |
| 5237 | Mastering the database environment |
| 2013 |
| 5243 | DATABASES in Theory |
| 2013 |

```

```

|      5248 | What You Always Wanted to Know About Database, But Were Afraid to
Ask |      2014 |
|      5252 | Beyond the Database Veil
|      2014 |
+-----+
+-----+
+-----+
5 rows in set (0.001 sec)

```

#12

```

MariaDB [FACT]> select check_num, book_num, check_out_date from checkout where
check_out_date < '2015-04-05';
+-----+-----+-----+
| check_num | book_num | check_out_date |
+-----+-----+-----+
|      91001 |      5235 | 2015-03-31 |
|      91002 |      5238 | 2015-03-31 |
|      91003 |      5240 | 2015-03-31 |
|      91004 |      5237 | 2015-03-31 |
|      91005 |      5236 | 2015-03-31 |
+-----+-----+-----+
5 rows in set (0.000 sec)

```

#13

```

MariaDB [FACT]> select book_num, book_title, book_year from book where book_year
> 2013 and book_title like '%programming%';
+-----+-----+-----+
| book_num | book_title | book_year |
+-----+-----+-----+
|      5247 | Shining Through the Cloud: Sun Programming |      2014 |
|      5253 | Virtual Programming for Virtual Environments |      2014 |
+-----+-----+-----+
2 rows in set (0.000 sec)

```

#14

```

MariaDB [FACT]> select book_num, book_title, book_year, book_subject, book_cost
from book where book_subject in ('cloud','middleware') and book_cost > 70;
+-----+-----+-----+-----+
| book_num | book_title | book_year | book_subject |
| book_cost |
+-----+-----+-----+-----+
|      5236 | Database in the Cloud |      2012 | Cloud |
|      79.95 |
|      5245 | The Golden Road to Platform independence |      2014 | Middleware |
|     119.95 |
|      5250 | Reengineering the Middle Tier |      2014 | Middleware |
|      89.95 |
+-----+-----+-----+-----+
+-----+-----+
3 rows in set (0.001 sec)

```

#15

```

MariaDB [FACT]> select au_id, au_fname, au_lname, au_birthyear from author where
au_birthyear between 1980 and 1989;
+-----+-----+-----+-----+
| au_id | au_fname | au_lname | au_birthyear |
+-----+-----+-----+-----+
|     218 | Rachel | Beatney |      1983 |
|     383 | Neal | Walsh |      1980 |

```

394	Robert	Lake	1982
438	Perry	Pearson	1986
460	Connie	Paulsen	1983
581	Manish	Aggerwal	1984
603	Julia	Palca	1988

+-----+-----+-----+

7 rows in set (0.000 sec)

#16

MariaDB [FACT]> select book_num, book_title, book_year

-> from book

-> where book_title like '%database%';

+-----+

+-----+

+-----+

book_num	book_title
book_year	

--

+-----+

+-----+

+-----+

5236	Database in the Cloud
------	-----------------------

2012

5237	Mastering the database environment
------	------------------------------------

2013

5243	DATABASES in Theory
------	---------------------

2013

5248	What You Always Wanted to Know About Database, But Were Afraid to
------	---

2014

5252	Beyond the Database Veil
------	--------------------------

2014

+-----+

+-----+

+-----+

5 rows in set (0.000 sec)

#17

MariaDB [FACT]> select pat_id, pat_fname, pat_lname

-> from patron

-> where pat_type = 'Student';

+-----+-----+-----+

pat_id	pat_fname	pat_lname

+-----+-----+-----+

1166	Vera	Alvarado
------	------	----------

1171	Peggy	Marsh
------	-------	-------

1172	Tony	Miles
------	------	-------

1174	Betsy	Malone
------	-------	--------

1180	Nadine	Blair
------	--------	-------

1181	Allen	Horne
------	-------	-------

1182	Jamal	Melendez
------	-------	----------

1184	Jimmie	Love
------	--------	------

1185	Sandra	Yang
------	--------	------

1200	Lorenzo	Torres
------	---------	--------

1201	Shelby	Noble
------	--------	-------

1202	Holly	Anthony
------	-------	---------

1203	Tyler	Pope
------	-------	------

1204	Thomas	Duran
------	--------	-------

1205	Claire	Gomez
------	--------	-------

1207	Iva	Ramos
------	-----	-------

1208	Ollie	Cantrell
------	-------	----------

1209	Rena	Mathis
------	------	--------

1210	Keith	Cooley
------	-------	--------

1211	Jerald	Gaines
1212	Iva	McClain
1213	Desiree	Rivas
1214	Marina	King
1215	Maureen	Downs
1218	Angel	Terrell
1219	Desiree	Harrington
1220	Carlton	Morton
1221	Gloria	Pitts
1222	Zach	Kelly
1223	Jose	Hays
1224	Jewel	England
1225	Wilfred	Fuller
1226	Jeff	Owens
1227	Alicia	Dickson
1228	Homer	Goodman
1229	Gerald	Burke
1237	Brandi	Larson
1238	Erika	Bowen
1239	Elton	Irwin
1240	Jan	Joyce
1241	Irene	West
1242	Mario	King
1243	Roberto	Kennedy
1244	Leon	Richmond

44 rows in set (0.000 sec)

#18

```
MariaDB [FACT]> select pat_id, pat_fname, pat_lname, pat_type
-> from patron
-> where pat_lname like 'c%';
```

pat_id	pat_fname	pat_lname	pat_type
1160	robert	carter	Faculty
1208	Ollie	Cantrell	Student
1210	Keith	Cooley	STUdent

3 rows in set (0.000 sec)

#19

```
MariaDB [FACT]> select au_id, au_fname, au_lname from author where au_birthyear
is NULL;
```

au_id	au_fname	au_lname
229	Carmine	Salvadore
262	Xia	Chiang
559	Rachel	McGill

3 rows in set (0.001 sec)

#20

```
MariaDB [FACT]> select au_id, au_fname, au_lname
-> from author
-> where au_birthyear is not NULL;
```

au_id	au_fname	au_lname
185	Benson	Reeves

218	Rachel	Beatney
251	Hugo	Bruer
273	Reba	Durante
284	Trina	Tankersly
383	Neal	Walsh
394	Robert	Lake
438	Perry	Pearson
460	Connie	Paulsen
581	Manish	Aggerwal
592	Lawrence	Sheel
603	Julia	Palca

12 rows in set (0.000 sec)

#21

MariaDB [FACT]> select check_num, book_num, pat_id, check_out_date, check_due_date from checkout where check_in_date is null order by book_num asc;

check_num	book_num	pat_id	check_out_date	check_due_date
91068	5238	1229	2015-05-24	2015-05-31
91053	5240	1212	2015-05-09	2015-05-16
91066	5242	1228	2015-05-19	2015-05-26
91061	5246	1172	2015-05-15	2015-05-22
91059	5249	1207	2015-05-10	2015-05-17
91067	5252	1229	2015-05-24	2015-05-31

6 rows in set (0.001 sec)

#22

MariaDB [FACT]> select au_id, au_fname, au_lname, au_birthyear
-> from author
-> order by au_birthyear desc,
-> au_lname asc;

au_id	au_fname	au_lname	au_birthyear
185	Benson	Reeves	1990
603	Julia	Palca	1988
438	Perry	Pearson	1986
581	Manish	Aggerwal	1984
218	Rachel	Beatney	1983
460	Connie	Paulsen	1983
394	Robert	Lake	1982
383	Neal	Walsh	1980
592	Lawrence	Sheel	1976
251	Hugo	Bruer	1972
273	Reba	Durante	1969
284	Trina	Tankersly	1961
262	Xia	Chiang	NULL
559	Rachel	McGill	NULL
229	Carmin	Salvadore	NULL

15 rows in set (0.000 sec)

#23

MariaDB [FACT]> select count(*) as 'Number of Books' from book;

Number of Books
20

```
+-----+
1 row in set (0.000 sec)
```

#24

```
MariaDB [FACT]> select count(distinct book_subject) as 'Number of Subjects' from
book;
```

```
+-----+
| Number of Subjects |
+-----+
|                    4 |
+-----+
1 row in set (0.000 sec)
```

#25

```
MariaDB [FACT]> SELECT COUNT(BOOK_NUM) AS "Available Books"
-> FROM BOOK
-> WHERE PAT_ID IS NULL;
```

```
+-----+
| Available Books |
+-----+
|                14 |
+-----+
1 row in set (0.000 sec)
```

#26

```
MariaDB [FACT]> select max(book_cost) as 'Most Expensive'
-> from book;
```

```
+-----+
| Most Expensive |
+-----+
|          129.95 |
+-----+
1 row in set (0.000 sec)
```

#27

```
MariaDB [FACT]> select min(book_cost) as 'Most Expensive' from book;
```

```
+-----+
| Most Expensive |
+-----+
|          49.95 |
+-----+
1 row in set (0.003 sec)
```

#28

```
MariaDB [FACT]> select count(distinct pat_id)as 'Different Patrons' from
checkout;
```

```
+-----+
| Different Patrons |
+-----+
|                 33 |
+-----+
1 row in set (0.000 sec)
```

#29

```
MariaDB [FACT]> select book_subject, count(book_subject) as 'Books in Subject'
from book group by book_subject order by count(book_subject) desc;
```

```
+-----+-----+
| book_subject | Books in Subject |
```

```

+-----+-----+
| Programming |          9 |
| Database    |          4 |
| Cloud       |          4 |
| Middleware  |          3 |
+-----+-----+
4 rows in set (0.002 sec)

```

#30

```

MariaDB [FACT]> select au_id, count(book_num) as 'Books Written'
-> from writes
-> group by au_id
-> order by count(book_num) desc,
-> au_id asc;

```

```

+-----+-----+
| au_id | Books Written |
+-----+-----+
| 262   | 3             |
| 460   | 3             |
| 185   | 2             |
| 229   | 2             |
| 251   | 2             |
| 383   | 2             |
| 394   | 2             |
| 559   | 2             |
| 218   | 1             |
| 273   | 1             |
| 284   | 1             |
| 438   | 1             |
| 581   | 1             |
| 592   | 1             |
| 603   | 1             |
+-----+-----+
15 rows in set (0.002 sec)

```

#31

```

MariaDB [FACT]> select sum(book_cost) as 'Library Value' from book;

```

```

+-----+
| Library Value |
+-----+
| 1499.00      |
+-----+

```

```

1 row in set (0.001 sec)

```